Faculty of Computer Applications

Introduction To Data Analytics

Smart City Management



Master of Computer Applications

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(2026-27)

DECLARATION

I, Sanskriti Saxena , hereby declare that the project entitled "Smart City Management System" submitted for the Master of Computer Applications (AI &ML) degree is my original work and has been carried out under the guidance of Mr. Divyank Chauhan, Assistant Professor, Department of Computer Applications.

The work embodied in this project has not been submitted elsewhere for the award of any other degree or diploma.

All sources of information have been duly acknowledged. The project work was completed as a collaborative effort with my team members, and each member contributed equally to its successful completion.

Team Members Contribution:

S. No.	Student Name	Contribution
1	Sanskriti Saxena (Leader)	
2	Niharika Negi	
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5	Vaishnavi Varshney	
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Signature of Team Leader

MCA Section A

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CERTIFICATE

This is to certify that the project entitled "Smart City Management System" is a bonafide work carried out by Sanskriti Saxena (Team Leader) along with the team members of MCA Section A, during the academic year 2026-2027, in partial fulfillment of the requirements for the award of Master of Computer Applications (MCA) degree.

The project work has been carried out under my guidance and supervision. The team demonstrated excellent teamwork, technical skills, and dedication throughout the project duration.

I am satisfied with the quality of work and recommend this project for evaluation.

_____

Mr. Divyank Chauhan

Assistant Professor

Project Guide

External Examiner: _____

Date: _____

Signature: _____

ACKNOWLEDGMENT

We would like to express our sincere gratitude to all those who have contributed to the successful completion of this project on "Smart City Management System."

First and foremost, we extend our heartfelt thanks to our esteemed guide, Mr. Divyank Chauhan, Assistant Professor, Department of Computer Applications, for his invaluable guidance, constant encouragement, and insightful suggestions throughout the project. His expertise and mentorship were instrumental in shaping this project.

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We are thankful to our family members and friends for their unwavering support and encouragement, which motivated us to put in our best efforts.

Last but not least, we acknowledge the collaborative spirit of our team members. Each team member contributed wholeheartedly, bringing diverse skills and perspectives that enriched the project. The synergy and coordination among the team made this project a rewarding learning experience.

We hope this project contributes to the field of healthcare data analytics and serves as a useful reference for future research.

Team Members: Sanskriti Saxena (Leader), Niharika Negi, Anmol Mishra, Priyanshi Dubey, Vaishnavi Varshney, Ansh Kulshrestha, Adama Ousman,Satyam, Alankar Maurya.

MCA(AI & ML)

ABSTRACT

Project Title: Smart City Management System
Domain: Smart City Analytics, Data Analytics, Business Intelligence
Technology Stack: Microsoft Excel, Power BI, Python, Jupyter Notebook, Data Visualization, Statistical Analysis

This project presents the development of a Smart City Management System using data analytics and interactive dashboards. The main objective of this project is to convert raw urban data into meaningful visual insights that support better decision-making and improve city management. With the rapid growth of urban populations, cities face several challenges related to crime, traffic congestion, energy consumption, and waste management. Traditional data management methods often lack analytical capabilities, making it difficult to identify patterns, monitor performance, and optimize urban resources.

To address these challenges, interactive dashboards were developed for four major domains: Crime Data Analysis, Traffic Analysis, Energy Management, and Waste Management. The dashboards were created using Microsoft Excel, Power BI, and Jupyter Notebook. Excel was used for data organization and interactive visualizations, Power BI for KPI cards, charts, filters, and interactive reporting, and Jupyter Notebook for data preprocessing, statistical analysis, and visualization using Python libraries. The dashboards provide insights into crime patterns, traffic conditions, energy consumption, and waste management activities, helping users identify trends and understand areas that require improvement.

The developed solution demonstrates how data analytics and business intelligence tools can be used to transform raw urban data into actionable insights. The outcome of this project highlights that interactive dashboards can reduce manual reporting efforts, improve data interpretation, support efficient resource allocation, and facilitate data-driven decision-making for smarter and more sustainable city management.

Keywords: Smart City Management, Crime Data Analysis, Traffic Analysis, Energy Management, Waste Management, Data Analytics, Business Intelligence, Excel, Power BI, Jupyter Notebook, Data Visualization, KPI.

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TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION

1.1 Background .....
1.2 Motivation .....
1.3 Problem Statement .....
1.4 Objectives .....
1.5 Scope of the Project .....
1.6 Project Overview .....

CHAPTER 2: LITERATURE REVIEW

2.1 Existing System Analysis .....
2.2 Research Gap .....

CHAPTER 3: METHODOLOGY

3.1 Data Collection .....
3.2 Data Cleaning & Preprocessing .....
3.3 Data Modeling (Pivot Tables / Power Query) .....
3.4 Dashboard Designing Process .....

CHAPTER 4: SYSTEM DESIGN

4.1 Tools & Technologies Used .....
4.2 ER / Data Flow Diagram .....
4.3 Architecture of Dashboard .....

CHAPTER 5: IMPLEMENTATION

5.1 KPI (Key Performance Indicator) Setup .....
5.2 Disease Distribution Chart .....
5.3 Age Group Distribution Chart .....
5.4 Gender Distribution Chart .....
5.5 Hospital Performance Chart .....
5.6 Cost & Critical Patient Analysis .....

CHAPTER 6: RESULTS & ANALYSIS

6.1 Dashboard Output Screenshots .....
6.2 Insights / Observations .....

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Conclusion .....
7.2 Limitations .....
7.3 Future Scope .....

REFERENCES .....

ANNEXURES / APPENDIX (Raw Data & Code) .....

CHAPTER 1: INTRODUCTION

1.1 Background

The rapid growth of urban populations has increased the need for efficient city management and sustainable development. Modern cities face several challenges, including crime, traffic congestion, increasing energy consumption, and inefficient waste management. Managing these challenges requires proper data collection, analysis, and monitoring to support effective decision-making. Traditional methods of managing urban information often depend on manual reporting and separate data systems, making it difficult to identify patterns and obtain meaningful insights.

The Smart City Management System is a data analytics-based project designed to analyze four major areas of urban management: Crime Data Analysis, Traffic Analysis, Energy Management, and Waste Management. The project uses Microsoft Excel, Power BI, Python, Jupyter Notebook, and database integration to process, analyze, and visualize urban datasets. Python supports data preprocessing and database connectivity, while Excel and Power BI are used to create interactive dashboards containing charts, tables, filters, and key performance indicators (KPIs).

The main purpose of this project is to transform raw data into understandable visual information that can help users identify trends, compare performance, and understand urban challenges. By integrating data analytics, database management, and visualization techniques, the project demonstrates how technology can support informed decision-making and contribute to more efficient and sustainable city management.

1.2 Motivation

The primary motivation behind developing the Smart City Management System is the increasing importance of data-driven decision-making in urban development. As cities grow, managing public safety, transportation, energy resources, and waste disposal becomes more complex. Analyzing these areas separately through traditional methods can be time-consuming and may limit the ability to identify important patterns and operational issues.

This project aims to simplify the analysis of urban datasets by presenting information through interactive dashboards. The use of Microsoft Excel and Power BI makes it easier to understand complex information through visual representations, while Python and database integration support organized data processing and storage. Jupyter Notebook provides an environment for exploratory data analysis and statistical investigation.

1.3 Problem Statement

Urban management involves handling large amounts of data related to crime, traffic conditions, energy consumption, and waste management. However, analyzing these datasets

using traditional methods can be challenging because of the volume of information, limited visualization capabilities, and the difficulty of identifying meaningful patterns through raw data alone.

Manual analysis and reporting can require considerable time and effort. Furthermore, when information is examined only in tabular form, understanding trends, comparing performance, and identifying areas requiring improvement may become difficult. Without appropriate analytical tools, it can be challenging to interpret urban data efficiently and use it to support informed decisions.

Therefore, there is a need for a system that can organize, analyze, and visualize urban datasets using suitable data analytics and business intelligence technologies. The Smart City Management System addresses this need by developing separate dashboards for Crime Data Analysis, Traffic Analysis, Energy Management, and Waste Management. Python is used for data processing and database connectivity, while Microsoft Excel, Power BI, and Jupyter Notebook support data analysis and visualization. The proposed system aims to simplify data interpretation, improve reporting efficiency, and provide useful insights for urban management.

1.4 Objectives

The main objective of the Smart City Management System is to develop a data analytics solution that transforms urban datasets into meaningful visual insights through data processing, database integration, and interactive dashboards.

The specific objectives of the project are:

1. To analyze crime-related data and identify patterns and trends using suitable visualizations.
2. To examine traffic conditions, vehicle movement, congestion, and other available traffic indicators.
3. To analyze energy consumption patterns and understand resource utilization.
4. To study waste management data and visualize relevant collection and operational trends.
5. To use Python for data preprocessing, analysis, and database connectivity.

1.5 Scope of the Project

The scope of the Smart City Management System focuses on analyzing and visualizing data from four major domains: crime, traffic, energy, and waste management. The project uses data analytics and business intelligence techniques to make urban information easier to understand and interpret.

The crime analysis component focuses on identifying crime patterns and distributions from the available dataset. The traffic analysis component examines indicators such as vehicle counts, traffic speed, road occupancy, and traffic conditions, depending on the available data. The energy management component analyzes energy consumption patterns, while the waste management component focuses on available information related to waste collection and management activities.

1.6 Project Overview

The Smart City Management System follows a systematic workflow involving data collection, preprocessing, database integration, analysis, and dashboard development. The project begins with datasets related to crime, traffic, energy consumption, and waste management. These datasets are prepared for analysis using Python and appropriate data processing techniques.

After preprocessing, the data is stored in or connected to a database using Python, supporting structured data management and retrieval. Jupyter Notebook is then used to explore the datasets, calculate relevant statistics, identify patterns, and generate visualizations. Microsoft Excel and Power BI are used to develop separate dashboards for the four domains.

Each dashboard presents relevant information through charts, tables, filters, and key performance indicators. These visual elements help users compare values, explore trends, and understand the characteristics of the available data. The dashboards developed across Excel, Power BI, and Jupyter Notebook also demonstrate different approaches to data analysis and visualization.

Overall, the project integrates programming, database management, data analysis, and business intelligence into a single analytical initiative. It demonstrates how urban datasets can be transformed into meaningful information that supports better understanding of city operations and provides a foundation for future smart city analytics solutions.

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CHAPTER 2: LITERATURE REVIEW

2.1 Existing System Analysis

Existing urban management systems use different methods to collect, store, analyze, and monitor city-related data. Information related to crime, traffic, energy consumption, and waste management is often maintained in separate datasets or information systems. Traditional approaches primarily focus on data storage and basic reporting, which may limit the ability to identify patterns and generate meaningful insights.

Microsoft Excel is widely used for organizing datasets, performing calculations, creating PivotTables, and developing charts. However, analyzing large datasets manually can become time-consuming. Power BI provides interactive dashboards, KPI cards, charts, and filters that help users explore information more effectively. Python, along with libraries such as Pandas, Matplotlib, and Seaborn, supports data preprocessing, statistical analysis, and visualization. Database integration further enables structured data storage and retrieval.

In the existing approach, each urban management domain may be analyzed independently. Crime data is examined to understand crime patterns, traffic data is used to study congestion and vehicle movement, energy data is analyzed to understand consumption patterns, and waste data is used to examine collection and management activities.

The Smart City Management System applies these technologies to four urban domains by developing separate dashboards using Excel, Power BI, and Jupyter Notebook. Python is used for data processing and database connectivity. This approach demonstrates how different analytical tools can convert raw datasets into understandable visual information and support more efficient data interpretation.

Although these tools provide useful analytical capabilities, the quality of the results depends on the accuracy, completeness, and availability of the underlying data. The developed project therefore focuses on data analysis and visualization rather than direct control of urban infrastructure.

2.2 Research Gap

Despite the availability of data analytics and business intelligence tools, several challenges remain in the analysis of urban management data. Information related to crime, traffic, energy consumption, and waste management is often analyzed separately, making it difficult to understand the different aspects of urban operations through a consistent analytical approach. Furthermore, raw datasets may contain missing values, duplicate records, inconsistent formats, and other data quality issues that affect analysis.

Traditional reporting methods may also lack interactive visualizations, making it difficult for users to filter information, compare categories, and identify important trends. Although tools such as Excel, Power BI, and Python offer powerful analytical capabilities, using them effectively requires appropriate data preprocessing, visualization techniques, and data management practices.

The major research gaps addressed by this project are:

1. **Fragmented Data Analysis:** Urban datasets related to crime, traffic, energy, and waste

management are often examined independently, limiting the understanding of individual urban challenges through a common project framework.

2. **Limited Interactive Visualization:** Basic reports and tabular data may not provide sufficient visual interaction for exploring trends, comparing categories, and understanding key performance indicators.
3. **Data Quality Challenges:** Inconsistent formats, missing values, and duplicate records can affect the accuracy and reliability of analytical results.
4. **Lack of Integrated Analytical Workflow:** Data preprocessing, database connectivity, statistical analysis, and dashboard development may be performed using separate tools without a clearly organized workflow.
5. **Difficulty in Comparing Analytical Tools:** Different visualization platforms offer different features, but their practical application to the same urban management domains may not always be explored together.

To address these gaps, the Smart City Management System uses Python for data processing and database connectivity, Jupyter Notebook for exploratory data analysis, and Microsoft Excel and Power BI for dashboard development. Separate dashboards are developed for crime, traffic, energy, and waste management to provide domain-specific insights through charts, KPIs, and other visual elements.

The project demonstrates a practical approach to organizing and visualizing urban data using multiple analytical technologies. It provides a foundation for future enhancements, including real-time data integration, predictive analytics, machine learning, and more advanced urban monitoring capabilities.

CHAPTER 3: METHODOLOGY

3.1 Data Collection

The first stage of the project involved collecting patient data. The dataset used in this project contains the following key attributes:

- Patient ID
- Name
- Age
- Gender
- Hospital Name
- City / Location
- Disease / Diagnosis
- Admission Status (Admitted, Discharged, Critical)
- Treatment Cost

The dataset was created and stored in Microsoft Excel. This data was either manually entered or imported from sample healthcare data sources. The purpose of data collection was to build a dataset that could represent real-world hospital and patient scenarios.

The data served as the foundation for the dashboard and further analytical processes.

3.2 Data Cleaning & Preprocessing

Once the dataset was collected, it was cleaned and processed to remove unnecessary or inconsistent entries. Data cleaning ensures accuracy and reliability of the dashboard insights.

Steps performed in data cleaning:

Step	Action Performed
Removal of Duplicates	Duplicate records were identified and removed to maintain data accuracy.
Handling Missing Values	Missing or blank values were checked and corrected where appropriate.
Data Formatting	Columns were formatted using appropriate data types, such as Text, Number, and Date.
Standardization	Inconsistent category values, such as disease names, hospital names, and gender values, were standardized for consistent analysis.

Examples:

- “M” and “Male” were made consistent (only “Male” used).
- Extra spaces and spelling errors were corrected.
- Age and cost values were converted to numeric format.

After preprocessing, the dataset became ready for analysis and dashboard creation.

3.3 Data Modeling (Pivot Tables / Power Query)

Data modeling involves structuring and organizing data so that Excel can analyze it efficiently.

In this project, **Pivot Tables** were primarily used for modeling and extracting insights.

Steps involved:

1. Creating multiple Pivot Tables from the dataset such as:
    - Gender-wise patient count
    - Disease category distribution
    - Age group classification
    - Hospital-wise performance
    - Patient status count (Admitted / Discharged / Critical)
1. Connecting these pivot tables with slicers for dynamic filtering.
2. Using Power Query (if required) to:
    - Combine data sheets
    - Remove errors
    - Ensure data consistency

The pivot tables act as the analytical engine of the dashboard.

3.1 Dashboard Designing Process

Once pivot tables were ready, visual elements (charts) were added to build the dashboard.

Steps followed in dashboard design:

Step	Description
Chart Selection	Selected appropriate charts, including Pie, Donut, Bar, and Column charts, to visualize data effectively.
KPI Display	Added key performance indicators (KPIs) to display important metrics such as Total Patients, Patient Ratio, and Critical Count.
Slicer Integration	Added interactive slicers to filter data by hospital, disease, gender, and city.
Layout Design	Arranged charts, KPI cards, and slicers in a clean, organized, and structured dashboard layout.

The dashboard was formatted to make it visually appealing, easy to interpret, and interactive.

Finally, the dashboard provided **real-time insights** and allowed users to filter data and view results instantly.

CHAPTER 4: SYSTEM DESIGN

4.1 Tools & Technologies Used

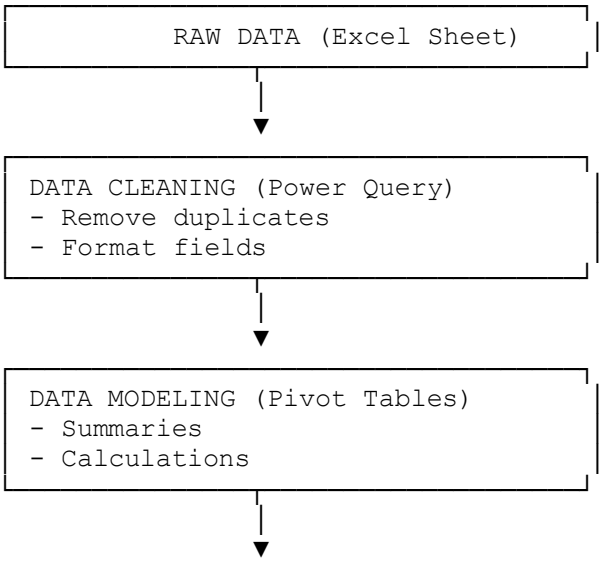
The Healthcare Dashboard project was developed using *Microsoft Excel* and its built-in analytical tools. These tools helped transform raw patient data into meaningful insights.

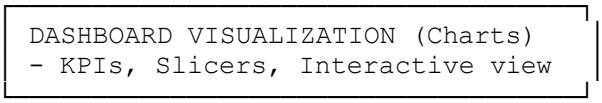
Tool / Technology	Purpose / Function
Microsoft Excel	Main platform used for data analysis and dashboard development.
Power Query	Cleans and preprocesses data by removing duplicates and transforming data types.
Pivot Tables	Used to summarize, group, and analyze data efficiently.
Charts (Column, Bar, Pie, Donut)	Convert analytical results into visual insights for better understanding.
Slicers / Filters	Enable users to interact with the dashboard and filter data dynamically.
Excel Formulas & Formatting	Used to calculate KPI values and enhance the visual appearance of the dashboard.

Excel was chosen because it is easy to use, widely available, and powerful for analytical dashboard creation **without coding**.

4.2 Data Flow Diagram (DFD)

The workflow of the project is based on converting raw data into visual analytics.
Improved diagram:





Process Explanation:

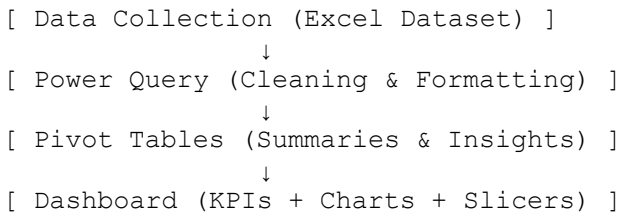
1. Raw dataset is imported into Excel.
2. Power Query cleans the data and prepares it for analysis.
3. Pivot tables summarize insights .
4. Dashboard charts and KPIs are created using pivot results.
5. Slicers allow users to interact with the dashboard dynamically.

4.3 Architecture of Dashboard

The architecture defines how components interact to convert raw data into a visual dashboard.

Layer	Component	Function
Data Layer	Excel Dataset	Stores raw data for analysis.
Processing Layer	Power Query and Pivot Tables	Cleans, transforms, and summarizes data for analysis.
Visualization Layer	Charts, KPI Cards, and Slicers	Converts analyzed data into interactive visual insights.

Dashboard Architecture Flow



This layered architecture ensures smooth data processing, easy updating, and real-time visualization.

CHAPTER 5: IMPLEMENTATION

CHAPTER 6: RESULTS & ANALYSIS

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Conclusion

The Smart City Management System was developed to analyze and visualize urban data across four major domains: Crime Data Analysis, Traffic Analysis, Energy Management, and Waste Management. The project demonstrates how data analytics and business intelligence tools can transform raw datasets into meaningful visual insights that support better understanding of urban challenges.

The project utilizes Microsoft Excel, Power BI, Python, Jupyter Notebook, and database integration to perform data preprocessing, analysis, storage, and visualization. Separate dashboards were developed for each domain to present relevant information through charts, tables, filters, and key performance indicators (KPIs). Python was used for data processing and database connectivity, while Jupyter Notebook supported exploratory data analysis and visualization. Excel and Power BI provided interactive ways to explore and interpret the analyzed data.

The developed dashboards help identify patterns in crime data, examine traffic conditions, understand energy consumption, and analyze waste management activities. The use of multiple analytical platforms also demonstrates different approaches to presenting data and generating insights. Database integration further supports structured data storage and retrieval.

Overall, this project provided practical experience in data analytics, database management, statistical analysis, and dashboard development. It demonstrates the importance of data-driven approaches in understanding urban operations and provides a foundation for developing more advanced smart city analytics solutions in the future.

7.2 Limitations

Although the Smart City Management System provides useful analytical visualizations, it has certain limitations.

1. **Data Availability:** The accuracy and usefulness of the analysis depend on the availability, completeness, and quality of the datasets used in the project.
2. **Data Quality:** Missing values, duplicate records, inconsistent formats, and incorrect entries may affect analytical results if they are not handled appropriately during preprocessing.
3. **Limited Real-Time Monitoring:** The dashboards primarily analyze the available datasets and do not necessarily provide live updates from city infrastructure or monitoring devices.
4. **Descriptive Analysis:** The project mainly focuses on data analysis and visualization. Advanced predictive models for forecasting crime patterns, traffic congestion, energy demand, or waste generation are not currently included.
5. **Separate Domain Dashboards:** Crime, traffic, energy, and waste management are analyzed through separate dashboards. The project does not provide a fully unified, real-time city monitoring platform.
6. **Dependence on Data Updates:** The usefulness of the dashboards depends on the availability of updated data and the implemented data-refresh process.

7. **Limited Operational Control:** The system is designed for analysis and decision support rather than directly controlling traffic signals, energy distribution, waste collection vehicles, or other urban infrastructure.

Despite these limitations, the project successfully demonstrates the practical application of data analytics, database connectivity, and visualization tools for understanding important urban management challenge

7.3 Future Scope

The Smart City Management System can be enhanced in several ways to improve its analytical capabilities and practical applications. Future developments could focus on real-time monitoring, predictive analytics, automation, and integration with smart city infrastructure.

1. **Real-Time Data Integration:** Live data from traffic sensors, surveillance systems, smart energy meters, and waste monitoring devices could be integrated to provide more frequently updated insights.
2. **Machine Learning Integration:** Machine learning models could be developed to predict crime trends, forecast traffic congestion, estimate energy demand, and predict waste generation based on historical data.
3. **Unified Smart City Dashboard:** The four domains could be integrated into a centralized dashboard to provide a consolidated overview of crime, traffic, energy, and waste management indicators.
4. **Automated Alerts:** The system could generate notifications when predefined conditions occur, such as unusual traffic congestion, increased energy consumption, or significant changes in crime-related indicators.
5. **Advanced Database Management:** Database connectivity could be enhanced through automated data pipelines, scheduled updates, and optimized queries to improve data accessibility and processing efficiency.
6. **Geographical Visualization:** Geographic Information Systems (GIS) and interactive maps could be incorporated to visualize crime locations, traffic hotspots, waste collection zones, and energy usage across different areas of a city.
7. **Web-Based Application Development:** The system could be extended into a web application that allows authorized users to access dashboards, explore analytical reports, and monitor relevant indicators from different devices.
8. **Sustainable Urban Planning:** Further analysis could help identify opportunities to optimize energy usage, improve waste collection planning, and support more efficient allocation of urban resources.

In conclusion, these enhancements could transform the current analytical project into a more comprehensive smart city decision-support platform. By incorporating real-time data, machine learning, automation, and integrated visualization, the system could provide more advanced insights to support efficient urban management and sustainable city development.

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REFERENCES

(You can list references you used during your project. Since your dashboard used *Excel* and concepts related to data visualization, here are standard references you can include.)

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ANNEXURES / APPENDIX